

SURAJ GUPTA GUDLA

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PROFESSIONAL SUMMARY

Results-driven Full-stack Software Development Engineer with 6+ years of experience building scalable, high-performance systems and cloud-native applications. Currently an SDE II at Amazon, specializing in distributed systems, AWS services, and end-to-end product development. Proven track record of architecting resilient micro-services, optimizing data pipelines, and delivering automation solutions that significantly reduce operational overhead and improve system reliability. Strong expertise in Java, Python, React, and AWS technologies, with a focus on performance optimization, CI/CD, and data-driven decision making. Adept at collaborating across cross-functional teams to deliver impactful solutions in fast-paced agile environments.

EDUCATION:

Master of Science in Computer Science

GPA: 3.7/4

Indiana University Bloomington - Bloomington, IN, USA

Jan 2021 - May 2022

Bachelor of Engineering in Electronics and Communication

GPA: 9.3/10

Chaitanya Bharathi Institute of Technology (Osmania University) - Hyderabad, India

Sep 2014 - May 2018

PROFESSIONAL EXPERIENCE: (6+ years)

Software Development Engineer 2

Dec 2024 – PRESENT

Amazon, Austin, TX

1. Architected and executed a zero-downtime migration of 100K+ site hierarchy records from DynamoDB to S3 to eliminate DynamoDB's 400 KB item size limitation that was causing write failures and threatening business continuity. Developed Kotlin-based migration utilities, refactored the data access layer to support S3-backed storage, enhanced serialization/deserialization frameworks, and implemented backward-compatible dual-read access using migration status tracking and S3 object references stored in DynamoDB. Built automated validation, backup, and rollback mechanisms, delivered phased deployments and controlled cutovers, and developed a new OpenSearch ingestion trigger to preserve free text search functionality. Successfully completed the migration with no downtime, no data loss, uninterrupted CRUD operations, and improved long-term scalability of the platform.
2. Built a scalable AI-powered product support chatbot using ReactJS, API Gateway, Lambda, Amazon Bedrock, and Bedrock Knowledge Bases, enabling business stakeholders to self-serve operational and troubleshooting queries without direct access to internal systems. Developed a custom MCP server exposing secure tools for DynamoDB, OpenSearch, CloudWatch, and internal APIs, allowing the LLM to combine knowledge-base context with real-time system data. Persisted chat history in DocumentDB for analytics and continuous improvement, significantly reducing developer support overhead and accelerating issue resolution.
3. Architected and built a scalable end-to-end data export service on AWS ECS using Kotlin, with auto-scaling to handle dynamic traffic. Integrated multiple services to aggregate and transform safety enforcement data, generate reports, and upload them to S3. Built AppSync GraphQL APIs and a React-based UI for report generation, historical retrieval, localization, and secure pre-signed downloads, enabling self-service access to operational data and eliminating manual reporting processes.
4. Designed and built a custom RAG-based semantic search platform for fulfillment center hazard & control data by developing an event-driven ingestion pipeline using DynamoDB Streams and Lambda, generating Titan embeddings through Amazon Bedrock, and indexing vectors in OpenSearch for KNN-based semantic retrieval. Built GraphQL APIs supporting natural-language search, server-side filtering, sorting, and pagination, replacing client-side processing and reducing load time by 10x.
5. Eliminated 60 days of developer operational effort annually by engineering a bulk site data validation tool using python featuring a dynamically configurable rule engine for 100+ key fields integrated with AWS AppConfig, enabling rapid rule updates without code deployment while automating site configuration change verification and empowering program & business teams to identify and correct data inconsistencies proactively.
6. Developed a high-performance migration automation tool in Kotlin to transfer 90K+ safety risk assessment records from a legacy system within three days. Implemented advanced capabilities including multi-threading,

local caching, rate limiting, batch processing, idempotent operations, robust exception handling, automatic rollbacks, live progress tracking, and automated credential refresh, leveraging AI-assisted development tools (Kiro, Cline) to accelerate delivery.

1. Reduced UI form template update time by 83% (from 30 to less than 5 minutes) by developing a generic automation tool in Kotlin which eliminates series of manual inputs and hard-coded values while supporting multi-stage, multi-program, multi-tenant updates.
1. Successfully on-boarded AWS data centers into the risk and safety product in 50% less time through strategic project management, parallel execution, and AI-driven optimization. Led refactoring and deployment of 4 front-end applications and 5 back-end micro-services, incorporating continuous stakeholder feedback through UAT cycles, demos, and discussions, while driving solution ideation, evaluation of alternatives, and feature prioritization.

Software Development Engineer 1

Jul 2022 – Nov 2024

Amazon Web Services, Seattle, WA

1. Designed and developed AWS Marketplace funding eligibility calculator micro-service in python supporting 180K+ clients with real-time consistency by integrating event-driven architecture to process live Salesforce updates; engineered LIST and GET REST APIs with server-side pagination and filtering capabilities, optimizing query performance through strategic DynamoDB GSI implementation across multiple fields and robust data schema design for sub-second response times.
2. Engineered end-to-end funding management solutions by developing backend services—including REST APIs, AWS Step Function workflows, REST APIs and event-driven handlers using AWS Lambda—to power reliable & accurate funding operations, claim transactions, and end-to-end workflow automation from request through credit disbursement seamlessly; while building intuitive frontend experiences with React-based web form interfaces to deliver a cohesive and user-friendly application.
3. Led the implementation of Strategic Collaboration Agreement user portal using React JS and AWS Cloudscape components, delivering intuitive experience and seamless functionality to AWS market-place partners.
4. Redesigned the opportunity selection interface, reducing user friction by 40% by eliminating unnecessary clicks and introducing a single-page flow. Improved usability with enhanced error handling and contextual help text, and optimized performance by replacing client-side filtering with server-side filtering, resulting in faster load times.
5. Transformed multi-stage deployment pipeline into full CI/CD by enhancing unit & integration test coverage, implementing production canaries, configuring bake-time and rollback monitors, establishing deployment failure notifications, enriching log details, emitting custom CloudWatch metrics, creating alarms, and developing service performance monitoring dashboards.
6. Provided accurate funding balance (cash + credits) insights via pie charts to marketplace partners by developing Python REST APIs integrated with Salesforce CRM via GraphQL, aggregating data from funding program databases and real-time currency conversion services, and optimizing performance through parallel API calls to deliver low-latency.
7. Developed end-to-end canary suite using Cypress JS to monitor availability, authentication, authorization, and critical business logic of the funding service by simulating artificial user traffic patterns and validating system reliability 24/7.
8. Implemented AWS CloudWatch RUM on the React-based funding benefits portal to capture client-side performance metrics, including user and page metadata, errors, and latency, improving frontend observability; and enhanced backend monitoring by introducing custom API metrics & alarms (error rates, latency, memory usage) along with robust logging and exception handling to ensure comprehensive server-side observability.

Software Engineer 1

Jul 2018 – Dec 2020

NCR Corporation, Hyderabad, India

1. Collaborated with cross-functional teams to deliver ATM software security enhancements, including remote OS hardening, locked-down user accounts, disk encryption, BIOS flashing and BIOS-level security using C# .NET Core and scripting tools, consistently meeting delivery timelines.
2. Reduced testing effort by 70% by automating test suites, configuration, and deployment workflows using AutoIT V3 scripts, significantly improving DevOps efficiency.
3. Migrated legacy AutoIT automation scripts to C# .NET, improving maintainability, code integrity, and extensibility of automation frameworks.
4. Managed Windows Server environments, SQL Server (SSMS), and Microsoft Active Directory (ADDS) to support application deployment and operational needs.

5. Proactively identified and resolved production and development issues with a turnaround time of 1–2 days, reducing security risks, bugs, code smells, and memory leaks through regular analysis using Coverity and SonarQube.

TECHNICAL SKILLS:

- **Programming & Scripting Languages:** Java, Kotlin, Python, Node.js, C#, Bash, PowerShell, Auto IT
- **Backend & API Development:** REST APIs, GraphQL APIs, Fast API, Spring, JPA, OOPS & Design Patterns, DSA, Junit, Pytest
- **Frontend & Web Technologies:** React.js, JavaScript, TypeScript, HTML, CSS, Cypress (Testing), Jest
- **Databases & Data Stores:** MS SQL Server, PostgreSQL, MongoDB, DynamoDB, Amazon DocumentDB
- **Cloud & Infrastructure (AWS):** Lambda, S3, SQS, SNS, DynamoDB, IAM, AppConfig, OpenSearch, ECS, AppSync, API Gateway, Cognito, CloudFormation, CloudTrail, Route 53, CloudWatch, AWS SDK
- **DevOps & CI/CD:** Docker, Kubernetes, Jenkins, CI/CD Pipelines, Git
- **Gen AI:** LLMs, Retrieval-Augmented Generation (RAG), Agentic AI, Model Context Protocol (MCP), Tooling & Function Calling, Prompt Engineering & Optimization, AI-Assisted Development